

- | Gas | Approximate percentage (%) of gases in the Earth's atmosphere | |
|----------------|---|-----------------------|
| | Original | Today |
| carbon dioxide | 75 | 0.04 |
| water vapour | 25 | variable, between 1-2 |
| oxygen | 0 | 21 |

Explain how the changes to these gases have taken place over geological time. [6 QER]

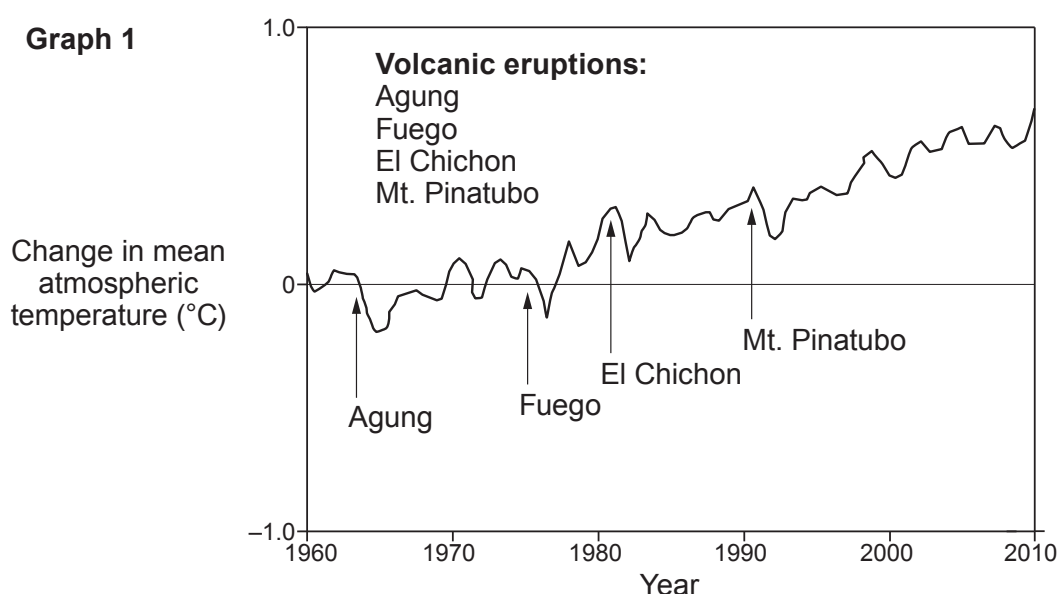
- (b) Use the information on this page to answer parts (i)–(iii).

Do Volcanoes Cause Global Warming?

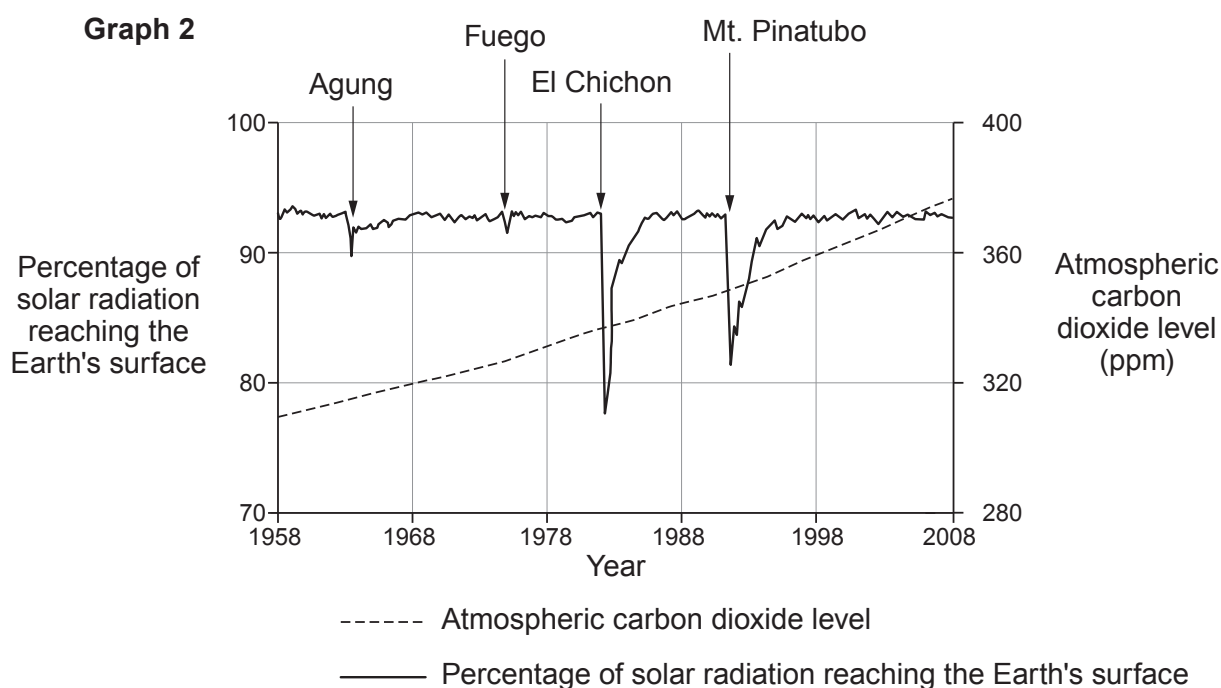
Volcanoes release vast amounts of greenhouse gases, such as carbon dioxide, which contribute to global warming. **Graph 1** shows the effect of four volcanic eruptions on the mean atmospheric temperature.

Erupting volcanoes also emit large quantities of sulfur dioxide into the atmosphere. Unlike carbon dioxide, sulfur dioxide increases the reflection of radiation from the Sun back into space, which cools the Earth's atmosphere. **Graph 2** shows the effect of the same four large volcanic eruptions on the percentage of solar radiation reaching the Earth's surface. The graph also shows the change in carbon dioxide levels in the atmosphere over the same time period.

Graph 1



Graph 2



- (i) Tick (✓) the box next to the statement which best describes the effect of volcanic eruptions on the overall level of carbon dioxide in the atmosphere. [1]

No significant impact on the overall level of carbon dioxide

☐

A significant increase in the level of carbon dioxide

☐

A significant decrease in the level of carbon dioxide

☐

- (ii) Tick (✓) the box next to the statement which best describes the effect that volcanic eruptions have on the mean atmospheric temperature. [1]

Mean atmospheric temperature decreases

☐

Mean atmospheric temperature increases

☐

No effect on the mean atmospheric temperature

☐

- (iii) Tick (✓) the box next to the statement which best explains the change in solar radiation reaching the Earth's surface after volcanic eruptions. [1]

Solar radiation decreases because it is reflected by sulfur dioxide

☐

Solar radiation increases because it is absorbed by carbon dioxide

☐

Solar radiation increases because it is absorbed by carbon dioxide and sulfur dioxide

☐

Solar radiation decreases because it reacts with sulfur dioxide forming sulfuric acid

☐
